

# Mastering Home Theater: The HDMI 2.2 Advantage

Elevating the Home Cinema Experience with Massive Bandwidth, Lossless Audio, and Seamless Synchronization.

HDMI 2.2 is the latest standard designed for the massive data requirements of ultra-high resolutions, offering massive bandwidth as a future-proof backbone for immersive entertainment.



## AUDIO EXCELLENCE & SYNC



### Cinema-Grade Lossless Audio

Supports uncompressed formats like Dolby Atmos and DTS:X for a true theater experience.



### Enhanced A/V Synchronization

Advanced synchronization technology ensures dialogue and action remain perfectly aligned.



### Multi-Device Synchronization

Enables seamless communication and control across consoles, players, and sound systems.



## FUTURE-READY PERFORMANCE



### 96 Gbps Bandwidth

Massive data throughput supports ultra-high resolutions and extremely high refresh rates.



### 10-Year Future-Proofing

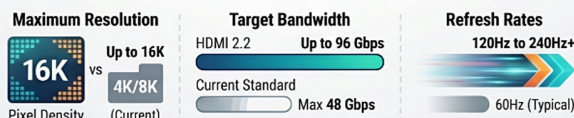
Ensures compatibility with upcoming 8K/16K broadcast ecosystems and next-gen gaming consoles.



### Content Ecosystem Readiness

Prepared for OTT platforms currently developing higher bitrate streaming formats.

## TECHNICAL CAPABILITIES COMPARISON



engineering milestone. Previous standards often required heavy chroma subsampling (reducing colour data) or restricted bit depth to fit within a smaller 'data pipe.' This substantial bandwidth expansion helps maintain signal integrity for uncompressed 8K at 120Hz, providing the overhead necessary for 10-bit and 12-bit HDR (high dynamic range) without losing fine detail to compression artefacts.

## Inherited and enhanced features: The technical core

HDMI 2.2 optimises the high-performance features introduced in version 2.1, refining them for ultra-high-bandwidth environments where signal timing is paramount:

### • Variable refresh rate (VRR).

Refined for seamless performance, allowing the display to match the source's frame rate dynamically, effectively eliminating screen tearing and stutter.

### • Auto low latency mode

(ALLM). Ensures the display automatically switches to its lowest-latency processing path when a gaming signal is detected, minimising the lag between input and

on-screen action.

• **Enhanced audio return channel (eARC).** Optimised to handle uncompressed, high-bitrate audio formats, eARC is essential for transmitting lossless object-based audio to external processors.

## The gaming paradigm: Performance and competitive edge

For the gaming sector, HDMI 2.2 offers a transformative shift in responsiveness. The standard is architected to meet the demands of high-end PC GPUs and future-generation consoles. By enabling 240Hz+ refresh rates, the interface provides a distinct competitive advantage in fast-paced gaming environments.

The combination of reduced latency and seamless VRR performance ensures smoother, more

responsive gameplay. For competitive gamers, this translates into a significant reduction in motion blur and input lag, ensuring the display never becomes the bottleneck for the user's reaction time.

## Home theatre excellence: Lossless audio and multi-device sync

Audiophiles and cinema enthusiasts benefit from the expanded bandwidth and improved synchronisation protocols of HDMI 2.2. The standard provides full support for lossless audio formats, including Dolby Atmos and DTS:X, ensuring the auditory experience is commensurate with 8K visual scales.

Furthermore, enhanced multi-device synchronisation enables a more cohesive home theatre ecosystem. This helps deliver a

## CORE SPECIFICATIONS

| Feature             | Specification                                         |
|---------------------|-------------------------------------------------------|
| Maximum resolution  | Up to 8K (uncompressed and compressed support)        |
| Refresh rates       | 120Hz native; potential for 240Hz+ via DSC            |
| Maximum bandwidth   | Up to ~96Gbps                                         |
| A/V synchronisation | Enhanced frame-level synchronisation protocols        |
| Media capabilities  | Advanced gaming and immersive broadcast media support |